

Auditing Large Language Models for Financial Trading Decisions

A Research Framework for Ground-Truth Evaluation, Calibration, Risk, Explainability and Trust

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Abstract

Large language models (LLMs) are increasingly used to interpret financial information, generate investment theses and support trading decisions. Their fluent explanations, however, do not establish predictive skill, calibration, financial consistency or risk adequacy. This paper develops a research-oriented framework for auditing LLM trading decisions using a structured Ground Truth (GT) table and a multi-dimensional evaluation layer. The framework records a model's recommendation and confidence at decision time, then joins it to future market outcomes such as 1-day, 5-day and 20-day returns, drawdown and risk-adjusted performance. It complements outcome evaluation with calibration, hallucination/evidence, financial-theory consistency, risk and explainability audits. Classical and deep-learning models—including XGBoost, LSTM and Transformer-style sequence models—serve as comparison baselines. The paper formalizes the mathematics behind returns, volatility, boosting, recurrent networks, self-attention, Sharpe/Sortino ratios, drawdown, VaR/CVaR, Brier score, Expected Calibration Error (ECE), SHAP/LIME and a composite Trust Score. Particular attention is given to time-series leakage, transaction costs, multiple testing and the distinction between engineering evidence, benchmark evidence and scientific claims. The resulting framework is best viewed as an AI model-risk and research-evaluation system, not as evidence that an LLM can reliably generate profitable trades.

1. Title & Research Positioning

Proposed research title: “Auditing Large Language Models for Financial Trading Decisions: A Ground-Truth, Statistical Calibration and Risk-Aware Evaluation Framework.”

The project is positioned at the intersection of quantitative finance, machine learning evaluation, financial model risk and generative AI. Its central contribution is methodological: rather than treating a convincing LLM explanation as evidence of trading skill, it creates an auditable chain from information available at time t , through model decision and confidence, to subsequently observable market outcomes.

This positioning is important because recent LLM-trading research has highlighted reproducibility problems involving data provenance, temporal splits, execution timing, turnover and transaction-cost assumptions. Related evaluation work has also shown that autonomous trading systems can be sensitive to perturbations in market intelligence, strategy, portfolio handling and execution.

Accordingly, the proposed framework treats the LLM as a decision-producing component inside a larger measurement system. The research objective is not to maximize a headline backtest return; it is to estimate whether the model's decisions are predictive, calibrated, financially coherent, risk-aware, robust and reproducible.

2. Research Problem

The research problem can be stated as follows: Given an LLM that observes information available at time t and outputs a trading recommendation a_t with confidence p_t , how can we determine whether the recommendation is economically useful and whether its stated confidence and rationale deserve to be trusted?

A binary accuracy metric is inadequate. A prediction may be directionally correct while producing poor risk-adjusted performance, excessive turnover or severe drawdown. Conversely, a low hit rate can coexist with positive expected value if winners are sufficiently large. Likewise, an LLM may achieve good returns while hallucinating evidence or being severely overconfident.

2.1 Failure modes to measure

- Outcome failure: the predicted direction or expected return does not match subsequent market behavior.
- Calibration failure: reported confidence is not aligned with empirical success probability.
- Evidence failure: factual financial claims are unsupported, stale or hallucinated.
- Theory failure: the narrative contradicts relationships implied by finance or by the supplied numerical data.
- Risk failure: the recommendation implies excessive exposure, drawdown or tail risk.
- Temporal leakage: the evaluation accidentally uses information unavailable at decision time.
- Execution failure: theoretical returns disappear after spread, slippage, fees, liquidity limits and latency.
- Selection failure: the strongest result is chosen from many trials without correcting for multiple testing.

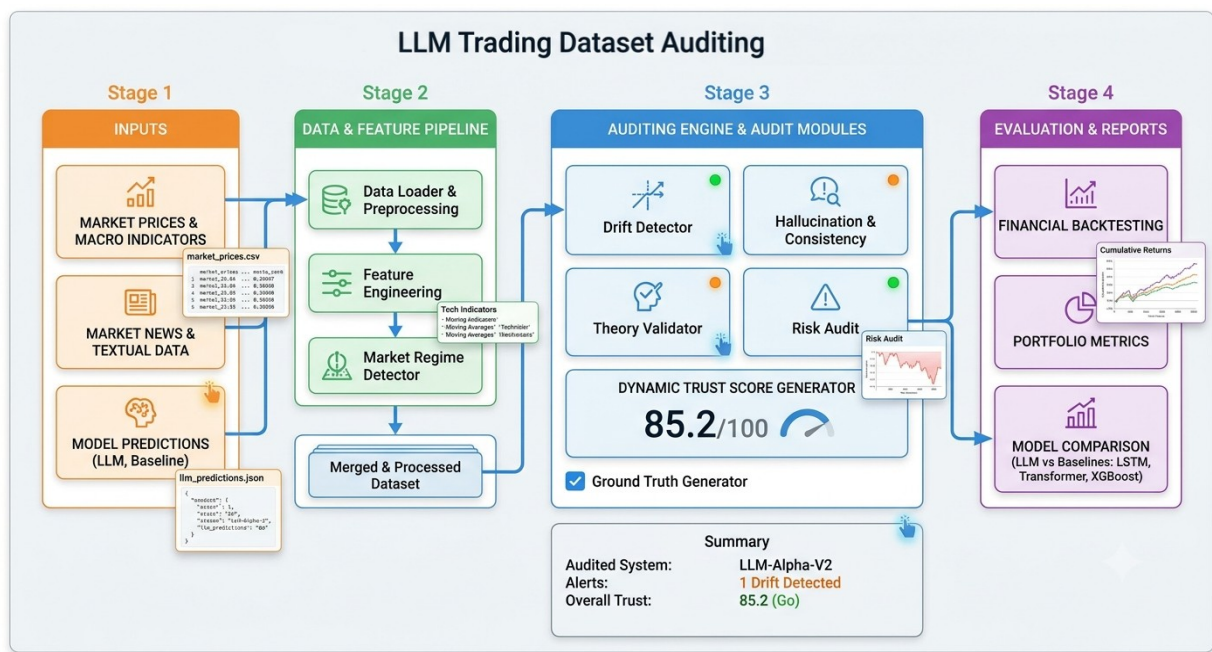
3. Research Objectives

- Construct a reproducible Ground Truth table linking every prediction to future realized market outcomes.
- Benchmark LLM decisions against classical ML and sequence-model baselines under a common protocol.
- Measure predictive performance at multiple horizons (1D, 5D and 20D).
- Quantify confidence calibration with Brier score, ECE and reliability analysis.
- Audit factual/evidential grounding and detect unsupported financial statements.
- Evaluate consistency with financial theory and numerical market features.
- Measure risk using volatility, Sharpe, Sortino, maximum drawdown, VaR and CVaR.
- Use SHAP/LIME-style explanations to inspect feature attribution for structured models.
- Develop a composite Trust Score while explicitly treating it as a model-risk indicator rather than a profitability guarantee.
- Define a rigorous experimental protocol using temporal validation, realistic costs and robustness tests.

4. End-to-End Architecture

The conceptual pipeline is:

Market data + news/macro information → preprocessing → feature engineering → model/LLM → decision + confidence → GT table → audit modules → Trust Score → research report/dashboard



4.1 Data layer

The data layer should contain point-in-time prices, volumes and any textual or macroeconomic information available before the decision timestamp. Every observation should carry a timestamp and provenance identifier. A point-in-time discipline is essential: adjusted data, revised fundamentals, future news or later corporate information must not leak backward into the decision set.

4.2 Modeling layer

The project can evaluate heterogeneous model families. XGBoost provides a nonlinear tabular baseline; LSTM provides recurrent temporal modeling; Transformer architectures provide attention-based sequence modeling; the LLM produces a natural-language investment decision and confidence. The key experimental requirement is that models receive comparable information and are evaluated on the same future horizons.

4.3 Audit layer

The audit layer transforms a prediction into a structured research record. Outcome, calibration, evidence, theory, risk, consistency and explainability modules should produce independent diagnostics before aggregation. This separation reduces the risk that one attractive metric masks another failure mode.

5. Ground Truth (GT) Table

The GT table is the central research artifact. Each row represents a decision made at time t . A conceptual schema is:

Field	Meaning	Example
prediction_id	Unique audit record	P000184
timestamp	Decision time	2026-05-12 10:00
ticker	Instrument	AAPL
model	LLM / XGBoost / LSTM / Transformer	LLM

action	BUY / SELL / HOLD	BUY
confidence	Reported probability/confidence	0.72
horizon	Evaluation horizon	5D
return_1d / 5d / 20d	Realized forward return	+2.1% / +4.3% / +1.2%
drawdown	Maximum drawdown over evaluation window	-3.4%
sharpe / sortino	Risk-adjusted outcome statistics	1.10 / 1.45
market_regime	Bull / Bear / Sideways / Crash	Bull
trust_score	Composite audit indicator	0.78

For a forward horizon h, the realized simple return is:

$$R(t, h) = \frac{P_{(t+h)}}{(P_t - 1)}$$

For a directional decision, a simple correctness indicator is:

$$C(t, h) = I[\text{sign}(\text{predicted direction}) = \text{sign}(R(t, h))]$$

This formulation makes the temporal boundary explicit: the model sees information available at t, while R(t,h) is computed only after h periods have elapsed.

6. Mathematical Model Foundations

6.1 Returns and volatility

Simple Return (r_t):

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Log Return (ℓ_t):

$$\ell_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns are additive across time and are frequently convenient for statistical modeling. A rolling volatility estimator is:

$$\sigma_t(n) = \sqrt{\left[\frac{1}{n-1} \sum_{i=0}^{n-1} (r_{t-i} - \bar{r}_t)^2 \right]}$$

6.2 Moving average and momentum

Moving Average ($MA_t(n)$):

$$MA_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

Momentum ($Momentum_t(k)$):

$$Momentum_t(k) = \frac{P_t}{P_{t-k}} - 1$$

These features provide structured numerical inputs against which an LLM-generated narrative can be checked. If a model claims “strong momentum” while the supplied momentum measure is negative, the discrepancy becomes an auditable consistency event.

7. Models Used and Mathematics

7.1 XGBoost / gradient-boosted trees

XGBoost represents a prediction as an additive ensemble of decision trees:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

At boosting step t , a regularized objective can be written conceptually as:

$$Obj^t = \sum_i L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t)$$

$$\Omega(f) = \gamma T + (1/2) \lambda \sum_{j=1}^T \omega_j^2$$

Here L is a loss function, T is the number of leaves and w represents leaf weights. The regularization discourages unnecessarily complex trees. In this project, XGBoost is valuable as a high-quality nonlinear tabular baseline rather than as proof that tree models are superior to LLMs.

7.2 LSTM

An LSTM models sequential dependence through gated state updates:

$$\begin{aligned} f_t &= \sigma(W_f[x_t, h_{t-1}] + b_f) \\ i_t &= \sigma(W_i[x_t, h_{t-1}] + b_i) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(W_c[x_t, h_{t-1}] + b_c) \\ h_t &= o_t \odot \tanh(c_t) \end{aligned}$$

The forget, input and output gates control what information is retained, written and exposed. This makes LSTM a natural baseline for price/feature sequences.

7.3 Transformer

Self-attention allows a sequence model to assign different weights to different positions:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The mechanism can capture long-range relationships without recurrent state updates. However, the same model can still suffer from temporal leakage if training and preprocessing are performed using future observations.

7.4 LLM decision model

The LLM is best treated as a probabilistic decision generator rather than a conventional calibrated classifier unless its probability estimates have been independently validated. A structured output should contain at minimum: action, confidence, thesis, evidence references, assumptions, expected horizon and invalidation conditions. Self-reported confidence should be audited rather than accepted as ground truth.

8. Evaluation & Quantitative Metrics

8.1 Return and wealth

$$W_T = W_0 \prod_{t=1}^T (1 + R_t)$$

The wealth curve is the basis for cumulative return and drawdown analysis. For implementable research, R_t should be net of fees and execution frictions.

8.2 Sharpe ratio

$$Sharpe = \frac{(E[R_p] - R_f)}{(\sigma_p \sqrt{A})}$$

A is the annualization factor. Sharpe rewards excess return per unit of total volatility.

8.3 Sortino ratio

$$Sortino = \frac{(E[R_p] - R_f)}{(\sigma_{down} \sqrt{A})}$$

Sortino replaces total volatility with downside deviation and is therefore useful when upside and downside variability should not be treated symmetrically.

8.4 Maximum drawdown

$$MDD = \max_t \left[\frac{(Peak_t - Wealth_t)}{Peak_t} \right]$$

Drawdown captures path-dependent loss severity. It is especially important for LLM systems because occasional concentration or overconfident decisions can create nonlinear portfolio damage.

8.5 VaR and CVaR

$$VaR_\alpha = -Quantile_\alpha(R)$$

$$CVaR_\alpha = -E[R | R \leq Quantile_\alpha(R)]$$

VaR summarizes a tail quantile while CVaR/Expected Shortfall estimates the average loss conditional on entering the tail. Both should be interpreted with care under non-normal returns.

8.6 Brier score

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

For binary events, $y_i \in \{0,1\}$. Lower is better. Brier score simultaneously rewards discrimination and probabilistic calibration.

8.7 Expected Calibration Error

$$ECE = \sum_b \left(\frac{n_b}{N} \right) |accuracy(b) - confidence(b)|$$

Predictions are grouped into confidence bins. ECE measures the weighted gap between average confidence and empirical accuracy. Reliability diagrams visualize the same relationship.

8.8 Information coefficient

$$IC_t = Corr(\text{predicted return}_t, \text{realized return}_t)$$

A time-series of cross-sectional or temporal IC values can test whether the model's ranking or return forecasts contain information beyond directional hit rate.

8.9 Transaction costs and execution

$$R_{net} = R_{gross} - fees - spread_{cost} - slippage - market_{impact}$$

This distinction is essential because a backtest that ignores implementation costs may report an economically irrelevant alpha.

9. Multi-Dimensional Audit Framework

The project is strongest when its audit dimensions remain separable before aggregation.

9.1 Prediction audit

Measures directional accuracy, return prediction quality, hit rate, IC and horizon-dependent outcomes. Evaluation should report confidence intervals, not only point estimates.

9.2 Calibration audit

Tests whether a stated confidence of 0.70 corresponds to roughly 70% empirical success in comparable cases. Overconfidence is a model-risk signal even if average returns are positive.

9.3 Hallucination / evidence audit

Each factual claim should be linked to a source, data field or deterministic calculation. A simple evidence-grounding ratio is:

$$EGS = \frac{(\text{supported claims})}{(\text{total auditable claims})}$$

More rigorous implementations should distinguish partially supported, stale, contradicted and fully supported claims.

9.4 Financial-theory audit

The system can check whether explanations are consistent with supplied numerical evidence and established relationships. Examples include CAPM beta interpretation, momentum, mean reversion, volatility and risk-return trade-offs. The purpose is not to force one theory of markets, but to detect internal contradictions.

9.5 Risk audit

Risk checks should consider position concentration, portfolio volatility, drawdown, leverage, liquidity, tail exposure and correlated positions. A trade-level BUY may be rejected at portfolio level if it materially increases an already concentrated factor exposure.

9.6 Explainability audit

For structured models, SHAP provides additive feature attribution:

$$f(x) = \varphi_0 + \sum \varphi_j$$

LIME constructs a local interpretable surrogate around an observation. These methods should be interpreted as explanations of the fitted model, not causal proofs.

9.7 Consistency and robustness audit

A model should be tested under changes in market regime, data perturbations, prompt variations and reasonable execution assumptions. Stability under small perturbations is a useful complement to average performance.

10. Composite Trust Score

A conceptual Trust Score can aggregate normalized audit dimensions:

$$T = w_1 Q_{pred} + w_2 Q_{cal} + w_3 Q_{evidence} + w_4 Q_{theory} + w_5 Q_{risk} + w_6 Q_{robustness}$$

The weights w_j should sum to one if the components are normalized to $[0,1]$. A weighted score is useful for dashboarding and triage, but it creates methodological risks: weights can be subjective, correlated metrics can be double-counted, and a single score can conceal catastrophic failure in one dimension.

Therefore, the recommended design is to publish both the Trust Score and the complete vector of component scores. A hard-risk gate can also prevent a high aggregate score from overriding a critical failure, e.g. severe leakage or unacceptable tail risk.

11. Experimental Design

11.1 Baseline protocol

- Freeze the universe, data sources, feature definitions, model versions and prompts before evaluation.
- Use chronological train/validation/test partitions.
- Use walk-forward evaluation for time-varying financial data.
- Ensure every feature is computable using only information available at the decision timestamp.
- Evaluate all models on identical prediction timestamps and horizons.
- Record random seeds, model versions, API/provider versions and preprocessing hashes where applicable.

11.2 Ablation studies

- LLM without news versus LLM with timestamp-safe news.

- Structured numerical features versus numerical + textual evidence.
- LLM confidence enabled versus confidence calibrated post hoc.
- Full audit versus individual audit modules.
- Gross returns versus net-of-cost returns.
- Single model versus multi-model ensemble.

11.3 Statistical inference

For each headline metric, estimate uncertainty using bootstrap or other appropriate time-series methods. Because observations can be serially dependent, naive independent-sample assumptions can understate uncertainty. Where strategies are compared repeatedly, multiple-testing corrections and methods such as Deflated Sharpe Ratio or Probability of Backtest Overfitting become relevant. Recent open research tooling explicitly emphasizes these issues.

11.4 Leakage and reproducibility

The strongest research design uses point-in-time data, explicit cutoff timestamps, no future-aware normalization, no random shuffling across time and an independently reproducible evaluation script. Recent LLM-finance research has specifically identified temporal discipline and execution assumptions as central reproducibility concerns.

12. Limitations

- An LLM's self-reported confidence is not automatically a statistically valid probability.
- Backtest performance does not establish live profitability.
- Small samples can make calibration and tail-risk estimates unstable.
- LLM provider/model updates can change behavior, reducing reproducibility.
- Textual evidence may be difficult to timestamp precisely and can contain revisions or survivorship effects.
- Composite Trust Scores depend on normalization and weighting choices.
- SHAP/LIME are explanatory approximations, not causal identification methods.
- Multiple experiments can generate false discoveries if the protocol is not pre-specified.
- Transaction-cost assumptions must be calibrated to the actual market, instrument and execution mechanism.
- Financial markets are non-stationary; a model can be calibrated in one regime and fail in another.

13. Research Extensions

13.1 Evidence graph

Represent each investment thesis as a graph connecting claims, evidence, numerical features, counter-evidence and final actions. This enables claim-level auditing instead of treating the entire explanation as one text object.

13.2 Adversarial red-team agents

Introduce a bearish/red-team agent whose objective is to falsify the thesis. Measure Decision Robustness:

$$DR = P(\text{decision remains unchanged} \mid \text{adversarial challenge})$$

13.3 Counterfactual testing

Perturb the evidence while holding other inputs fixed. A rational system should respond directionally to material changes in earnings growth, volatility, valuation or macro assumptions. This provides a sensitivity test for LLM decision logic.

13.4 Portfolio-level auditing

Move beyond individual trades to factor exposure, correlation, concentration, liquidity and portfolio optimization. Classical portfolio baselines such as equal weight, buy-and-hold, risk parity, minimum variance and Markowitz/MVO provide useful anchors in a rigorous benchmark.

13.5 Execution realism

Model latency, partial fills, spread, market impact, participation constraints and slippage. Recent LLM trading research emphasizes that execution assumptions can materially affect whether reported results are economically interpretable.

13.6 Pre-registration and independent replication

A high-quality scientific extension should freeze the primary hypothesis and analysis before inspecting final test results. This reduces researcher degrees of freedom and makes a null result scientifically valuable. Recent LLM trading work has adopted pre-registration specifically to separate a credible test from post-hoc storytelling.

14. Expected Research Contribution

The project contributes a unified measurement perspective in which predictive performance is only one component of model reliability. Its key contribution is the GT-table abstraction: every AI decision becomes a reproducible object that can be linked to future returns, risk outcomes, confidence calibration and qualitative audit findings.

The framework also clarifies an important distinction: a system may be profitable in a historical simulation yet fail as a trustworthy financial AI system because its confidence is miscalibrated, its explanations are unsupported, its performance disappears after costs, or its results depend on leakage. Conversely, an LLM may provide useful contextual information even if it is not an independent alpha generator. Contemporary benchmark projects increasingly make this distinction between trading skill and calibrated assistance explicit.

15. Conclusion & Future Direction

This research framework treats LLM trading as a model-risk problem rather than merely a stock-picking problem. The core methodology is to record decisions at time t , construct future Ground Truth outcomes, evaluate predictive and risk-adjusted performance, test probabilistic calibration, audit evidence and financial consistency, and expose the complete result through a reproducible audit trail.

The mathematical layer provides the quantitative foundation: returns and volatility describe market behavior; XGBoost, LSTM and Transformer equations define baseline model classes; Sharpe, Sortino, drawdown, VaR and CVaR quantify risk; Brier score and ECE quantify calibration; SHAP/LIME support model interpretation; and a composite Trust Score summarizes—but does not replace—the individual diagnostics.

The future research direction is an autonomous but human-gated quant research platform: evidence acquisition → multi-model research → thesis generation → adversarial review → portfolio/risk engine → realistic execution simulation → Ground Truth → statistical audit → model promotion decision. The most important scientific principle is that stronger claims require stronger evidence. The system should therefore be designed to make failure, uncertainty and non-replication visible rather than optimizing solely for attractive back test numbers.

References & Further Reading

- Project repository: github.com/macbhaldar/llm-trading-audit.
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- Yan, L., et al. (2025). TradeTrap: Are LLM-based Trading Agents Truly Reliable and Faithful? arXiv:2512.02261.
- Bailey, D. H., & López de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality.
- Aisha Alansari. (2026). Large Language Models Hallucination: A Comprehensive Survey. arXiv:2510.06265
- Related benchmark direction: TradeArena / TreLLM emphasizes replayable trajectories, risk gates, execution calibration and reproducible evidence.
- Related calibration evidence: contemporary finance research has documented overconfidence issues in LLM-based measures and the sensitivity of results to confidence extraction methods.